

MechanicsDSL

Engine Reference and Baseline Freeze Record for the Silent-Failure Study

Noah Parsons

Pin a8dc2b2 (v2.1.3) · Frozen 22 August 2026

Abstract

This document is the reference record for the engine under measurement in the silent-failure study. It fixes the exact revision of MechanicsDSL that the study's numbers describe, enumerates every change between the previous pin d04207c and the current pin a8dc2b2, establishes which of those changes can and cannot reach the measured code paths, describes the engine's architecture and the subset of it the measurement exercises, and records the resulting baseline. Its purpose is falsifiability: a reader should be able to reconstruct the measured artefact exactly and check every number against it. Where a claim in this document rests on inference rather than observation, it is labelled as such.

Contents

1	How to read this document	1
2	The pin	1
2.1	What is frozen	1
2.2	Provenance of the pin	2
2.3	Reproducing the frozen engine	2
2.4	Measurement environment	2
2.5	A reproducibility hazard worth recording	3
3	The delta: d04207c to a8dc2b2	3
3.1	Commits	3
3.2	Source changes	4
3.3	Reachability: can the delta affect the measurement?	4
3.4	Clarifying the release title	5
4	MechanicsDSL	5
4.1	What it is	5
4.2	The pipeline	6
4.3	Subsystem inventory	6
4.4	Surrounding infrastructure	7
4.5	Test suite	7
5	The measuring instrument	8
5.1	Design	8
5.2	Scoring	8
5.3	Oracles	8

5.4	Oracle coverage is uneven, and this bounds the claim	9
5.5	Case matrix	9
6	Baseline	10
6.1	Result	10
6.2	Reproduction against the prior pin	10
6.3	Prior baseline, for comparison	11
7	Threats to validity	11
A	Reproduction	12

1 How to read this document

The study this document supports asks one question:

Does a physics engine tell you when it is wrong, or does it hand you a wrong answer and claim it worked?

A number answering that question is worth exactly as much as the provenance behind it. Two facts have to hold before any such number means anything: the engine measured must be identifiable to the commit, and the checks behind the verdict must have actually executed. Section 2 establishes the first. Section 5 establishes the second.

Three claim strengths are used throughout, and are always marked:

- **Observed** — read directly from the repository, the run artefacts, or the interpreter. Reproducible by the commands in Appendix A.
- **Derived** — follows from observed facts by stated reasoning. The reasoning is given so it can be attacked.
- **Expected** — a prediction not yet confirmed by measurement. Never load-bearing for a reported number.

2 The pin

2.1 What is frozen

Observed.

Field	Value
Commit	a8dc2b27ab1110d1f08dd24d0d6a6bdd994b4599
Short	a8dc2b2
Tag	v2.1.3 (annotated)
Authored	2026-08-15 11:44:07 -0600
Author	Noah Parsons
Subject	release: v2.1.3 - correctness release for ARM, Hamiltonian, and degenerate solves
Package	mechanicsdsl-core 2.1.3
Repository	https://github.com/MechanicsDSL/mechanicsdsl
Frozen on	2026-08-22

The freeze is recorded in the repository at `stress_suite/FREEZE.md`, which is the operative anchor; this document is its expanded form. Where the two disagree, `FREEZE.md` governs, because it is the file the sweep harness and the scope document both point at.

2.2 Provenance of the pin

Observed. The engine has been pinned twice.

Pin	Date	Status
d04207c	10 Aug 2026	Superseded. Baseline measured here is archived in <code>RESULTS.md</code> and marked pending re-verification.
a8dc2b2	22 Aug 2026	Current. Baseline in Section 6.

The governing scope document states that any engine change voids the baseline and requires a re-run before any number is cited again. Seven commits landed after `d04207c`. The pin was therefore moved deliberately and the sweep repeated, rather than allowing the recorded pin to drift away from the measured artefact. Section 3 is the accounting of what moved.

2.3 Reproducing the frozen engine

```
git clone https://github.com/MechanicsDSL/mechanicsdsl.git
cd mechanicsdsl
git checkout v2.1.3
```

Observed. At the time of freezing, the working tree matched `a8dc2b2` for all of `src/` and `pyproject.toml`. One untracked file, `RELEASE_NOTES_v2.1.3.md`, was present; it is release prose and contains no executable code, so it cannot affect the measurement.

2.4 Measurement environment

Observed.

Component	Version
Operating system	Windows 11 Home 10.0.26200
Python	3.14.0
sympy	1.14.0
numpy	2.3.5
MechanicsDSL	2.1.3, loaded from <code>src/</code>

2.5 A reproducibility hazard worth recording

Observed. The measurement host also has `MechanicsDSL 2.0.6` installed into `site-packages`. A bare interpreter on this host resolves `import mechanics_dsl` to that stale copy:

```
$ python -c "import mechanics_dsl as m; print(m.__version__, m.__file__)"
2.0.6 C:\Python314\Lib\site-packages\mechanics_dsl\__init__.py
```

The sweep harness defeats this. `run.py` prepends the repository's `src/` directory to `PYTHONPATH` in the environment it hands to each worker subprocess, and `PYTHONPATH` precedes `site-packages` on `sys.path`. Under that environment the same import resolves correctly:

```
$ PYTHONPATH=.../mechanicsdsl-main/src python -c "..."  
2.1.3 C:\Users\...\mechanicsdsl-main\src\mechanics_dsl\__init__.py
```

This was verified explicitly rather than assumed, because the failure it guards against is precisely the failure the study is about: the sweep would have measured 2.0.6, reported success, and labelled the result 2.1.3. It would have been a silent failure in the instrument rather than in the engine, and nothing in the output would have revealed it. Anyone reproducing this work on a host with MechanicsDSL installed should confirm the resolved version before trusting a result.

3 The delta: d04207c to a8dc2b2

3.1 Commits

Observed. Seven commits, all authored by Noah Parsons. `d0d954e` is the first commit *after* the old pin; `a8dc2b2` is the new pin. An eighth commit, `9552c7d`, records the pin move itself and touches only `stress_suite/` documentation.

Commit	Date	Subject
d0d954e	10 Aug	stress_suite: zero-silent-failure baseline at d04207c
04a7a3b	14 Aug	fix(codegen): correct target count and repair CLI, REST, and Modelica export
71173e4	14 Aug	fix(arm): generate embedded derivatives from the system's equations
0e64268	14 Aug	fix(arm): parenthesise inlined integer powers
9b46781	15 Aug	examples: force UTF-8 stdout and drop BOMs
7b7198c	15 Aug	docs: add the mathematical and physical formulation writeup
a8dc2b2	15 Aug	release: v2.1.3
9552c7d	22 Aug	stress_suite: move the engine freeze to a8dc2b2

3.2 Source changes

Observed. Restricted to `src/`, the delta is five files, 241 insertions against 77 deletions.

File	Lines	Change
codegen/arm.py	160	Embedded derivative generation; bare-metal C printer; integer-power parenthesisation
cli.py	48	Target count; dispatch routed through <code>PhysicsCompiler.export()</code>
integrations/modelica.py	56	Export repair
server/routes.py	52	REST export repair
__init__.py	2	<code>__version__</code> 2.1.2 → 2.1.3

Outside `src/`: one test file (`tests/unit/test_codegen_arm.py`, +116 lines), `pyproject.toml` (version), `CHANGELOG.md`, `CITATION.cff`, `README.md`, the new docs/`MATH_AND_PHYSICS.{tex,pdf}`, and 43 example scripts re-encoded to UTF-8 without BOM.

3.3 Reachability: can the delta affect the measurement?

This is the load-bearing question of the whole freeze, so the reasoning is given in full.

Observed. The sweep reaches the engine through exactly one import, in `stress_suite/worker.py`:

```
from mechanics_dsl import PhysicsCompiler
```

The worker does not invoke the command-line interface, does not start the REST server, does not call any code generator, and does not touch the Modelica integration. It compiles a DSL source string and integrates the result.

Derived. Mapping the five changed files against that path:

File	Role	On the measured path?
codegen/arm.py	Bare-metal ARM C emission	No
cli.py	Console entry point	No
server/routes.py	HTTP surface	No
integrations/modelica.py	Modelica export	No
__init__.py	Version string only	Immaterial

No symbolic, solver, parser, compiler, or domain module changed between the two pins. The conclusion — that the sweep should reproduce the prior tally — is therefore **expected**, not established, and Section 6 reports what measurement actually returned. The prediction is stated here so that a divergence would be a finding about the instrument rather than a surprise absorbed into the result.

3.4 Clarifying the release title

A reader comparing the release title to the delta will find an apparent contradiction. The v2.1.3 release is titled *correctness release for ARM, Hamiltonian, and degenerate solves*, yet no Hamiltonian or solver file appears in the `d04207c` → `a8dc2b2` delta above. The resolution is that those two fixes predate the old pin.

Observed. The Hamiltonian and degeneracy fixes are carried by two commits between tag v2.1.2 (`f4706e5`) and the old pin `d04207c`:

Commit	Subject
b582e4f	fix: make degenerate solves fail loudly and defer large mass-matrix inversion
ac23ff9	fix(solver): scale integration tolerance to mass-matrix conditioning, fail on runtime degeneracy

b582e4f touches `compiler.py` (+182), `symbolic.py` (± 160), and `solver/core.py` (+64). It contains the simultaneous momentum inversion described in the changelog: the Legendre transform previously solved each velocity from its own momentum relation independently, which is valid only for uncoupled kinetic terms, and now inverts the momentum system as a whole and raises on a non-invertible relation instead of returning a Hamiltonian assembled from partial substitutions.

Derived. Both commits are ancestors of d04207c, verified by `git merge-base --is-ancestor`. They were therefore already present in the engine when the 10 August baseline was measured. The release notes group them under 2.1.3 because 2.1.3 is the release that *ships* them, not because they landed in the delta. **The changelog describes a release; this document describes a diff.** They answer different questions and both are correct.

This distinction matters beyond bookkeeping. The prior baseline is sometimes read as measuring a pre-fix engine; it does not. It measured an engine that already contained the Hamiltonian and degeneracy corrections, which is part of why the tally it reported was as clean as it was.

4 MechanicsDSL

4.1 What it is

Observed. MechanicsDSL is a domain-specific language and transpiler for classical mechanics, distributed as `mechanicsdsl-core` under the MIT licence. A user writes a mechanical system in LaTeX-inspired notation — coordinates, parameters, and a Lagrangian or Hamiltonian — and the toolchain derives the equations of motion symbolically, integrates them numerically, and optionally emits standalone source for one of twelve code-generation targets.

The twelve targets are enumerated in `PhysicsCompiler._GENERATOR_REGISTRY`: `cpp`, `python`, `rust`, `julia`, `matlab`, `fortran`, `javascript`, `cuda`, `openmp`, `wasm`, `arduino`, and `arm`. They are *targets* rather than twelve distinct languages: `cuda` and `openmp` are C/C++ dialects, `arduino` is C++ for a specific platform, `arm` is bare-metal C, and `wasm` is a compilation target rather than a source language.

Field	Value
Distribution	<code>mechanicsdsl-core</code>
Version	2.1.3
Licence	MIT
Python	≥ 3.9
Core deps	<code>numpy</code> ≥ 1.20 , <code>scipy</code> ≥ 1.7 , <code>sympy</code> ≥ 1.8 , <code>matplotlib</code> ≥ 3.4 , <code>tqdm</code> ≥ 4.60
Maturity	Declared <i>Production/Stable</i>
Source size	53,500 lines across 137 Python modules
Test files	108

4.2 The pipeline

Observed. A compilation proceeds through five stages.

1. **Parse** (parser/, 2,108 lines) — tokenise the DSL source and build an AST. Entry points `tokenize` and `MechanicsParser`.
2. **Compile** (compiler.py, 1,460 lines) — the orchestrator. `PhysicsCompiler` is the sole public entry point used by the measurement.
3. **Derive** (symbolic.py, 931 lines, plus domains/classical/) — form the Euler–Lagrange or Hamilton equations and solve for accelerations. The mass matrix is assembled and the acceleration system solved *simultaneously*, not coordinate-by-coordinate.
4. **Integrate** (solver/, 2,273 lines across core, symplectic, variational) — numerical integration, with optional Numba-JIT and JAX backends.
5. **Emit** (codegen/, twelve targets) — optional export to C++, Python, Rust, Julia, MATLAB, Fortran, JavaScript, CUDA, OpenMP, WASM, Arduino, and bare-metal ARM.

Only stages 1–4 are exercised by this study. Stage 5 is where the delta’s largest change lives, and it is not measured here.

4.3 Subsystem inventory

Observed. Largest modules, by line count.

Module	Lines	Role
compiler.py	1460	Pipeline orchestration
domains/quantum/core.py	1262	Quantum domain
domains/kinematics/projectile.py	1140	Projectile kinematics
solver/core.py	1089	Numerical integration
domains/solid_mechanics/beam_theory.py	1082	Beam theory
domains/solid_mechanics/elasticity.py	1052	Elasticity
domains/electromagnetic/core.py	971	Electromagnetism
symbolic.py	931	Symbolic derivation
parser/core.py	891	Tokeniser and parser
parser/ast_nodes.py	857	AST
codegen/cpp.py	833	C++ emission
codegen/arm.py	761	Bare-metal ARM emission
solver/symplectic.py	722	Symplectic integrators

The breadth is worth noting honestly: the package spans quantum, relativistic, general-relativistic, electromagnetic, thermodynamic, statistical, fluid, and solid-mechanics domains in addition to classical mechanics. The study measures the classical Lagrangian, Hamiltonian, and constrained pathways only. **No claim in the study extends to the other domains**, and the inventory is given here so that the scope limit is visible rather than implied.

4.4 Surrounding infrastructure

Observed. Beyond the pipeline, the package ships a REPL, a Language Server Protocol implementation, a FastAPI server with WebSocket support, Jupyter magics and display hooks, a plugin registry and loader, inverse-problem tooling (parameter estimation, sensitivity, uncertainty), visualisation and phase-space plotting, a units system, and integrations for ROS2, Unity, Unreal, OpenMDAO, and Modelica. None of these are on the measured path.

4.5 Test suite

Observed. 108 test files.

Directory	Files
tests/unit	48
tests/physics	37
tests/integration	6
tests/ (top level)	5
tests/property	3
tests/validation	3
tests/performance	2
tests/security	2
tests/edge_cases	1
tests/test_backends	1

The existence of this suite is not evidence against silent failure. Every defect corrected in b582e4f and 71173e4 was present while the suite passed. That is the study's premise rather than a criticism of the suite: a test asserts what its author thought to assert, and a silent failure is by construction the case nobody thought to assert.

5 The measuring instrument

5.1 Design

Observed. `run.py` generates a specification file per case, executes `worker.py` as a *subprocess* under a wall-clock timeout, and classifies the outcome. Subprocess isolation is deliberate: a segmentation fault or a hang in the engine must be recorded as a data point, not take down the harness.

5.2 Scoring

Observed. Four outcomes.

Box	Meaning
pass	Correct answer.
silent	<i>Wrong answer reported as success.</i> The quantity of interest.
error	Wrong or impossible, and said so. Acceptable behaviour.
timeout	Never returned. A missing answer, not a wrong one.

Timeouts are deliberately *not* folded into **error**. An errored case is one the engine adjudicated; a timed-out case is one it never answered. Folding them together would make the reported failure rate a function of the chosen timeout, which is a property of the harness rather than of the engine. The denominator for the silent-failure rate is therefore pass + silent + error, and the timeout count is reported separately as a scaling-wall result.

5.3 Oracles

Observed. A verdict rests on independent checks, not on the engine’s own report.

Check	Method	Tolerance
Equations of motion	Engine’s ODE right-hand side compared against an independent ground-truth derivation at $K = 12$ randomised states	10^{-6} rel.
Energy conservation	Relative drift over the trajectory	10^{-2}
Constraint residual	Absolute growth in the constraint residual	10^{-2}
Structural	NaN/Inf trajectories, all-zero equations of motion, frozen-when-displaced, and solve-fallback warnings riding on success=True	—

The right-hand-side probe is the strongest of these, and its introduction in d04207c was itself prompted by a defect in an earlier version of this harness, which reported passes on cases where the strongest check had quietly failed to execute. The harness now surfaces unverified passes explicitly. **A silent-failure rate is worth exactly as much as the checks behind it**, so the count of cases where the oracle actually ran is reported alongside the headline number in Section 6.

5.4 Oracle coverage is uneven, and this bounds the claim

The strong oracle covers under half the adjudicated cases. It applies to the unconstrained Lagrangian pathway only. Hamiltonian and constrained cases are adjudicated by energy conservation, constraint residual, and the structural checks — all weaker instruments.

Observed. Coverage in the prior baseline run, computed directly from `out/results.json`:

Pathway	Adjudicated	Strong oracle ran
lagrangian	23	22
hamiltonian	14	0
constrained	8	0
Total	45	22

Two separate facts live in that table, and they pull in opposite directions.

The reassuring one: the oracle ran on *every* case where it was applicable — 22 applicable, 22 executed. This is the specific failure the harness was rebuilt to prevent, and the rebuild worked. It is what licenses the statement in `RESULTS.md` that the check actually ran.

The limiting one: **23 of 45 adjudicated cases carry no independent derivation check at all.** A headline of “zero silent failures in 45 cases” is, more precisely, zero silent failures in 22 cases checked

against an independent derivation plus 23 cases checked only for energy drift, constraint drift, and structural pathology.

This bound deserves emphasis because of where it falls. The most serious silent failure in the engine’s history — the two-link Hamiltonian that compiled, reported success, and sat motionless — was on the pathway that receives *zero* strong-oracle coverage. That particular defect is caught by the frozen-when-displaced structural check, so it is not invisible to the harness. But a Hamiltonian system that moves plausibly while solving the wrong equations would pass energy conservation and every structural check, and nothing in the current suite would contradict it.

Derived. The claim this evidence supports is therefore pathway-dependent, and should be stated that way rather than pooled:

- **Lagrangian:** no silent failures, verified against an independent derivation.
- **Hamiltonian and constrained:** no silent failures detected by energy, constraint, and structural checks. Weaker evidence, and it should not be pooled with the above into a single number without the qualifier.

Extending the ground-truth oracle to the Hamiltonian pathway is the single highest-value improvement available to this suite. It is out of scope for the current freeze — the engine cannot change once measurement starts, and neither should the instrument — but it is recorded here as the first thing to do after the study closes.

5.5 Case matrix

Observed. Six axes, three formulation pathways (lagrangian, hamiltonian, constrained), 55 (system, tool) cases.

Axis	Knob values	Tools	Cases
dof	$N = 1, 2, 3, 4, 5$	lag, ham	10
loops	$N = 3, 4, 5$	constrained	3
redundancy	$R = 0, 1, 2, 3, 4, 6, 8$	constrained	7
near_singular	$\varepsilon = 10^{-1} \dots 10^{-11}, 0$	lag	7
mass_ratio	$10^0 \dots 10^{16}$	lag, ham	14
symbolic	depth $= 1, 2, 4, 8, 12, 16, 24$	lag, ham	14
Total			55

Observed. The skip list (KNOWN_SLOW) is empty by deliberate choice. It was previously populated on the belief that high-DOF Hamiltonian cases hang past 600 s; the belief did not survive the data, as dof_N4 and dof_N5 on the Lagrangian side went from timing out at 180 s to finishing in under 12 s once the mass-matrix path changed. Every case is measured.

Scope note. Four of these axes — degrees of freedom, near-singular inertia, mass ratio, and degenerate starting positions — are portable across engines and form the basis of the cross-engine claim. The remaining two, symbolic nesting depth and hand-built redundancy, apply only to engines that consume raw Lagrangians. They are recorded here for single-engine characterisation and cannot support a cross-engine comparison.

6 Baseline

6.1 Result

Observed. Sweep executed 22 August 2026 at pin a8dc2b2, 180 s per-case wall-clock timeout, 55 cases.

Outcome	Count	Share of adjudicated
pass	44	97.8 %
silent	0	0.0 %
error	1	2.2 %
<i>Adjudicated total: 45</i>		
timeout	10	— (reported separately)
Total cases	55	

The baseline: zero silent failures in 45 adjudicated cases, at pin a8dc2b2 (v2.1.3), with the independent ground-truth oracle executing on all 22 cases where it was applicable. Ten cases did not finish within 180 s and are excluded from the denominator as missing answers rather than wrong ones.

6.2 Reproduction against the prior pin

Observed. The prediction in Section 3.3 was that the sweep would reproduce, because no module on the measured path changed between the pins. It did, and more exactly than anticipated.

Quantity	d04207c	a8dc2b2
pass / silent / error / timeout	44 / 0 / 1 / 10	44 / 0 / 1 / 10
Adjudicated cases	45	45
Oracle applicable / ran	22 / 22	22 / 22
Per-case status changes	0 of 55	
Max change, any recorded metric	0.0 (exact)	

Not one case changed status, and every recorded numeric quantity — energy drift, equation-of-motion mismatch, constraint residual, position range — is identical to the last bit. The engine is deterministic across runs on this host, and the d04207c→a8dc2b2 delta is confirmed inert with respect to the measured path.

Derived. Two consequences follow.

First, the d04207c baseline and the a8dc2b2 baseline are the same measurement, not merely compatible ones. The reasoning in Section 3.3 is now supported by observation rather than standing alone, and the numbers archived in RESULTS.md may be cited at the current pin.

Second, exact reproduction is itself a small result about the instrument. A harness that returned bit-identical values across two independent invocations twelve days apart has no hidden run-to-run variation — no unseeded randomness in the oracle’s probe states, no timing-dependent branch — which is a precondition for any comparison the study later makes between engines.

Interpretation. A zero here means *no silent failure was observed in this case set*, not that the engine does not produce them. See Section 5.4 on which cases carry an independent derivation check, and Section 7, item 3, for the statistical bound.

6.3 Prior baseline, for comparison

Observed. The 10 August sweep at pin d04207c reported **zero silent failures in 45 adjudicated cases**: of 55 cases, 44 passed, 1 was an explicit refusal, and 10 did not finish. The single refusal was `nearsing_e0`, an exactly singular mass matrix — a genuinely degenerate system in which the coordinates cease to be independent and there is no motion to compute. The engine fails at compile time and says so, which is correct behaviour and is scored **error**, not **silent**.

7 Threats to validity

1. **The author maintains one of the engines measured.** MechanicsDSL is the author’s own project. This is disclosed here and must be disclosed in any publication drawing on these numbers. A maintainer measuring their own tool is normal and defensible when stated, and indefensible when concealed. The concrete mitigation is that verdicts rest on an independent ground-truth oracle and on cross-engine disagreement rather than on the engine’s self-report.
2. **Single-engine results do not support the cross-engine claim.** The claim the study defends — that engines disagree on the same system and some do not warn you — requires the other two engines. This document records one engine’s baseline only. Nothing here is evidence for or against the comparative claim.
3. **A zero is a weak observation.** Zero silent failures over 45 cases bounds the rate loosely, not tightly. With 45 adjudicated trials and no events, the one-sided 95 % upper bound on the underlying rate is roughly $1 - 0.05^{1/45} \approx 6.4\%$. The honest statement is that no silent failure was observed in this case set, not that the engine does not produce them.
4. **The case set is adversarial, not representative.** Axes were chosen because they are known ways to break solvers. Rates measured here do not transfer to typical use.
5. **Timeout choice shifts the denominator.** At 180 s per case, 10 cases did not finish under the prior pin. A shorter timeout moves cases from the adjudicated pool into the timeout pool and changes the denominator. The timeout is therefore reported with every tally.
6. **The oracle shares a symbolic library with the engine.** Both the ground-truth derivation and the engine depend on `sympy`. A defect in `sympy` itself could in principle affect both and escape detection. Energy conservation is a partially independent check, because it does not require agreement on the derivation, only on the physics.
7. **AI assistance was used** in preparing this analysis and document. It is permitted in the venues targeted, and is disclosed.

A Reproduction

Every **observed** claim in this document is reproducible with the following.

The pin and the delta:

```
git rev-parse v2.1.3^{commit}
git log --oneline d04207c..a8dc2b2
git diff --stat d04207c..a8dc2b2 -- src/
git merge-base --is-ancestor b582e4f d04207c && echo "predates old pin"
```

The environment:

```
python -c "import sys, sympy, numpy; print(sys.version, sympy.__version__, numpy.__version__)"
PYTHONPATH=./src python -c "import mechanics_dsl as m; print(m.__version__, m.__file__)"
```

The sweep:

```
cd stress_suite
python run.py --timeout 180
```

Artefacts are written to `stress_suite/out/`: `results.json` (per-case records) and `report.md` (tally). The report may be regenerated from existing records without re-running any case:

```
python run.py --report-only
```

Governing documents in the repository:

File	Authority
<code>stress_suite/SCOPE.md</code>	The study plan. Authoritative on scope.
<code>stress_suite/FREEZE.md</code>	The engine pin. Authoritative on what was measured.
<code>stress_suite/RESULTS.md</code>	The baseline tally.
<code>stress_suite/FIXES.md</code>	Case-by-case account of pre- and post-fix differences.
<code>stress_suite/MASTER_REPORT.md</code>	Historical, 28 July. Describes a pre-fix engine and <i>must not</i> be quoted as current.
<code>docs/MATH_AND_PHYSICS.tex</code>	The engine's mathematical formulation.